

Modelling of spatially correlated weather-based electricity forecasting using combined frequency-based signal decomposition with optimized boosting approach

Indra A. Aditya^a, Didit Adytia^{b,*}

^a Generation Division, PLN Research Institute, Jakarta, 12760, Indonesia

^b School of Computing, Telkom University, Bandung, 40257, Indonesia

ARTICLE INFO

Keywords:

Electricity load forecasting
Weather-based forecasting
Signal decomposition
Xgboost

ABSTRACT

Effective planning of electric energy systems is becoming increasingly vital due to the growing integration of renewable resources and the electrification of end-user industries. However, short-term electrical load forecasting continues to be challenging due to the effects of weather unpredictability and consumer behaviour. This research presents a machine learning-driven forecasting framework that incorporates spatially correlated meteorological data, temporal characteristics, and frequency-based signal decomposition using the Fourier Transform. The primary contribution is a spatially correlation-driven feature selection technique to choose ideal weather input sites, coupled with the extraction of predominant frequency components from the load signal to enhance model input. Three machine learning models are evaluated: XGBoost, AdaBoost, and Multi-Layer Perceptron (MLP) on datasets from two locations in Indonesia: Bali and Jakarta-Banten. XGBoost attained optimal performance with the five most frequent components. For Bali, the model produced an R^2 of 0.89, a correlation coefficient (CC) of 0.98, and a root mean square error (RMSE) of 37.83; for Jakarta-Banten, it gave an R^2 of 0.90, a CC of 0.95, and an RMSE of 497.99. These findings underscore the advantages of integrating spatial weather relevance with signal decomposition to improve prediction accuracy, which is essential for reliable and efficient power system operations.

1. Introduction

Despite its long-standing role as the primary energy source, human dependence on fossil fuels faces a critical challenge due to their finite availability, compounded by environmental repercussions [1]. In the realm of electricity generation, there is a notable transition underway, moving from fossil fuels towards renewable energy sources [2]. Despite the recognized benefits of renewable energies, their complete implementation is hindered by various barriers, including cost considerations, market share dynamics, and policy constraints [3]. Hence, the necessity for streamlined and efficient planning in electrical energy generation becomes paramount, aimed at mitigating wastage and enhancing overall efficiency. Short-term electricity load forecasting (STLF) is a fundamental facilitator of planning, essential for production scheduling, demand response, reserve allocation, and real-time system balancing [4,5]. However, the task of electricity forecasting is far from straightforward as it is subject to many intricate variables that can significantly impact its precision. Factors ranging from the dynamic nature of customer behaviour to the unpredictable vagaries of

weather conditions within the forecasted region introduce complexity and uncertainty into the forecasting process.

In forecasting methodologies, two prominent approaches emerge: statistical intelligence and artificial intelligence [6]. In a study conducted by Haiges et al. in 2017, [7], the forecasting of electricity generation capacity is explored by employing statistical methods, specifically autoregressive integrated moving average (ARIMA). Dudek in 2016 [8] employed local linear regression models based on patterns for short-term load forecasting. Meanwhile, Bouktif et al. in 2018 [9] investigates electricity load forecasting, utilizing the currently trending method of artificial intelligence, particularly emphasizing the use of deep learning long short-term memory (LSTM) [10] models. Yang et al. in 2019 [11] applied a machine learning approach known as the Least Square Support Vector Machine to predict short-term electricity load. The application of machine learning in electricity load forecasting was also explored in a paper by S. Li et al. [12]. In a study conducted by Zhang et al. in 2018 [13], a hybrid model was employed for electricity load forecasting. This model incorporates empirical mode decomposition

* Corresponding author.

E-mail address: adytia@telkomuniversity.ac.id (D. Adytia).

<https://doi.org/10.1016/j.ijepes.2025.110698>

Received 7 November 2024; Received in revised form 23 March 2025; Accepted 23 April 2025

Available online 16 May 2025

0142-0615/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

(EMD) [14], ARIMA [15], and wavelet neural network (WNN) [16], which are optimized using the fruit fly optimization algorithm (FOA).

Ahmad et al. [17] have introduced an enhanced short-term load forecasting (STLF) model, TFTformer, a transformer-based architecture that utilizes temporal embeddings and gated layers, surpassing traditional deep learning models in several domains. Liu et al. [18] devised a Temporal Depthwise Convolutional Network (TDCN) that enhances accuracy via dilated convolutions and depth filters. Dong et al. [19] employed a BiGRU-based stacking autoencoder that integrates bidirectional memory and attention mechanisms to enhance predicting in scenarios with limited data. Spatial characteristics are being examined with growing frequency. Zhao et al. [20] developed a transformer that incorporates graph attention to represent multivariate and spatial load interactions in residential forecasting. Gao et al. [32] illustrated the application of transfer learning and multi-attention fusion for predicting electric vehicle load, resulting in a substantial enhancement of accuracy in a sparse data context.

Nevertheless, a notable challenge in electricity forecasting lies in the modelling of features to achieve accurate prediction results. Weather is identified as a significant factor influencing electricity load [4]. However, given the vast coverage area, the optimal criteria for selecting weather sites remains uncertain. This creates a deficiency in contemporary research, i.e., the majority of present studies fail to systematically incorporate regional weather-load correlations and do not employ frequency-based signal decomposition for feature enhancement. Signal decomposition techniques have found extensive application across many research studies and have proven to be quite influential in improving the accuracy of a system. In a research paper by Arvanitidis et al. in 2022 [21], a comprehensive comparison was conducted between signal decomposition and clustering techniques for electricity price forecasting. In the context of short-term electricity price prediction, employing the Variational Mode Decomposition (VMD) method, which leverages signal decomposition, yielded the most accurate outcomes, with a remarkably low mean absolute percentage error value of 6.15%. This finding underscores the superior predictive performance of signal decomposition compared to the clustering-based approach. Zhou et al. in 2022 [22] undertook short-term photovoltaic power forecasting using a combination of signal decomposition and machine learning optimization techniques.

This study proposes a spatially-informed forecasting paradigm that merges weather-load correlation analysis with signal decomposition utilizing Fourier Transform to overcome the existing gap. We propose a technique for selecting weather features based on their correlation coefficient (CC) with electricity load. This involves calculating the CC of each weather feature with the electricity load. We also introduce a novel method for feature selection from weather data, which we refer to as Spatial Correlation. This method focuses on determining the optimal locations for weather data collection to enhance the accuracy of the predictions. Additionally, we present a signal decomposition technique utilizing the Fourier transform, along with proposed criteria for signal decomposition.

The machine learning models utilized comprise XGBoost [23], renowned for its exceptional prediction accuracy in gradient boosting, AdaBoost [24], another boosting algorithm, and Multi-layer Perceptron (MLP) [25]. For the test case, we selected two locations with distinct characteristics: Bali and the Jakarta-Banten area. Bali was chosen due to its closed geographical area, while the Jakarta-Banten area was selected for its complexity and highest electricity consumption.

Our research illustrates the novel utilization of Fourier decomposition and spatial feature selection to enhance load forecasting accuracy. While Fourier signal decomposition is an established technique, the novelty of our approach lies in its strategic use to refine forecasting models. Our research meticulously outlines the process of selecting key frequency components from the spectrum domain—specifically, the 5, 10, 15, and 24 most influential signals. These selected frequencies play a critical role in improving the model's predictive performance.

The true innovation of this work is not just in applying Fourier decomposition but in the detailed spectrum analysis and the informed selection of specific frequencies, which we argue is crucial for achieving higher forecasting precision. This methodical and sophisticated strategy enhances the accuracy and dependability of energy demand estimates, impacting actual power system operations.

The structure of this paper unfolds as follows. In Section 2, we delve into the realms of electricity forecasting, signal decomposition techniques, and their applications, and expound upon the theory underpinning boosting. Section 3 is dedicated to elucidating our proposed electricity load forecasting model. Section 4 is reserved for the presentation and discussion of the research findings. Lastly, Section 5 offers a concise summary of our work and outlines potential avenues for future research.

2. Literature review

Precise electrical load estimates are crucial for power system operators to guarantee reliable supply, enhance operational efficiency, and minimize generation costs. In recent years, the advancement of intelligent forecasting systems has gained significance due to the rising volatility of demand and the expansion of renewable energy resources. This paper presents a machine learning framework for short-term power load forecasting that combines boosting models, i.e., XGBoost, AdaBoost, and Multi-Layer Perceptron (MLP), with frequency-domain data decomposition using Fourier Transform. The literature is examined through two primary themes: electricity forecasting models and the utilization of signal decomposition in power system prediction.

2.1. Electricity forecasting

The precision of electricity forecasting predictions is essential for enabling electricity suppliers to align their production with consumer demand accurately. This alignment not only ensures a reliable supply but also aids electricity companies in minimizing operational costs.

Electricity forecasting, while not a novel concept, has evolved significantly over time, driven by the need to meet consumer demands efficiently and economically. Many methodologies have been proposed to enhance the accuracy of an electricity forecasting system. A study conducted by Felice et al. in 2013 [26] focused on forecasting electricity demand within Italy by leveraging numerical weather prediction (NWP) models. Dehalwar et al. in 2016 [27] undertook the development of an Artificial Neural Network (ANN) and Bagged Regression Trees to produce load forecasts in an urban setting, leveraging meteorological data for prediction purposes. Friedrich et al. in 2015 [28] carried out short-term electricity load forecasting for Abu Dhabi, employing a Transfer Function (TF) model that incorporated a range of meteorological variables as inputs. A research by Son and Kim in 2017 [29] was centred on short-term electricity demand forecasting within the residential sector of Korea. This study harnessed the predictive capabilities of Support Vector Machine (SVM) [14] modelling with comprehensive incorporation of meteorological and social variables to enhance accuracy. In a study by Elamin and Fukushima in 2018 [30], the forecasting of hourly load data of electricity demand was accomplished through the application of (seasonal autoregressive integrated and moving average with exogenous variable) SARIMAX with interaction.

Although these older methods provided significant contributions, subsequent research has progressively transitioned towards deep learning and hybrid methodologies. Ahmad et al. [17] announced TFTformer, a transformer-based architecture developed for short-term load forecasting with temporal feature embedding. Liu et al. [18] introduced a TDCN model employing extended depthwise convolution to enhance sequence learning, whilst Dong et al. [19] created a BiGRU-based stacked autoencoder with self-attention to capture intricate temporal

dynamics. Zhao et al. [20] and Gao et al. [5] emphasized the significance of spatial and spatial features through graph attention and transfer learning techniques for housing and electric vehicle forecasting applications. Notwithstanding these advancements, feature selection continues to provide a considerable problem, particularly when integrating meteorological data. Many models continue to handle weather input locations either randomly or based on proximity, lacking clear correlation analysis. Furthermore, the application of signal processing techniques, including signal decomposition, remains inadequately employed in enhancement-based forecasting frameworks, hence justifying the contribution of this paper.

2.2. Signal decomposition

A pivotal strategy for advancing prediction systems' accuracy entails applying signal decomposition. This proven technique, widely employed across various studies, consistently contributes to substantial improvements in predictive accuracy.

In a study conducted by Wang et al. in 2018 [31], the combination of SVM models and Signal Decomposition was harnessed to predict hourly solar radiation levels. Then, in a study by Feng et al. in 2022 [32], the integration of SVM and Signal Decomposition emerged again, this time employed for the modelling of hydrological forecasts. In the domain of electricity demand forecasting, An et al. in 2013 [33] leveraged a multi-output feedforward neural network in conjunction with signal filtering based on empirical mode decomposition. Jaseena and Kovoov in 2021 [34] combined a decomposition technique with a deep-learning BiLSTM model to effectively forecast wind speed. Subsequently, an innovative method was employed by da Silva et al. in 2021 [35], which melded Complete Ensemble Empirical Mode Decomposition (CEEMD) with Stacking-ensemble learning (STACK) driven by Machine Learning algorithms. This approach was deployed for wind energy forecasting within a wind farm in Parazinho City, Brazil, adopting a multi-step-ahead forecasting strategy.

As seen in the examples above, various ways exist for signal decomposition, and in this study, we employ the Fourier Transform for signal decomposing. Details regarding this approach will be expounded upon in the subsequent discussion.

2.3. Signal decomposition using Fourier transform

According to Zhang et al. [36], the Fourier Transform is a mathematical operation that breaks down a signal into its underlying frequencies. The original signal is associated with time and is known as the signal's time domain representation. In contrast, the Fourier Transform deals with frequency and is called the signal's frequency domain representation. The term "Fourier Transform" encompasses representing the signal in the frequency domain and the mathematical procedure used to achieve this transformation.

Eq. (1) demonstrates the application of the Fast Fourier Transform (FFT) in computing the Fourier Transform [37].

$$X(k) = \sum_{n=0}^{N-1} x(n)e^{-2\pi jnk/N} \quad (1)$$

$X(k)$ stands for the outcome derived from the Discrete Fourier Transform (DFT) at a particular frequency index, denoted by k , within the sequence. The original sequence of values, labelled as $x(n)$, serves as the input for the DFT process. The variable N signifies the total number of data points or samples present in the input sequence utilized for the DFT computation. The range of k spans from 0 to $N - 1$, encompassing all frequency indices in the input sequence for the DFT computation. Additionally, j represents the imaginary unit $\sqrt{-1}$.

In this study, Fourier transformation will be employed to analyse the electrical load signal, transforming it into the frequency domain to observe its spectrum. Through this analysis, we aim to identify the dominant frequency components within the electrical load signal. Subsequently, the most prominent frequency will be utilized for signal reconstruction.

2.4. Boosting

Boosting stands out as one of the most captivating ensemble learning techniques, capturing significant attention from researchers in the field of machine learning [38].

According to Zhou [39], boosting comprises a set of algorithms that empower weak learners to become robust predictors. These algorithms commence by training a base learner, subsequently modifying the distribution of training samples based on the base learner's outcomes. This adjustment prioritizes incorrectly classified samples for subsequent base learners. Following the training of the initial base learner, subsequent iterations involve training additional base learners with these adjusted samples, perpetuating the process until a specified number of base learners, denoted as T , is reached. Ultimately, these base learners are weighted and combined to form the boosted ensemble.

Among the array of boosting algorithms, AdaBoost stands out as one of the most widely recognized [39]. This boosting methodology evolved into gradient boosting, with XGBoost emerging as a prominent member within this framework [40,41].

2.5. Extreme Gradient Boosting

Extreme Gradient Boosting (XGBoost), proposed by Dr. Tianqi Chen, is recognized as a proficient and scalable implementation of the GBM (Gradient Boosting Method) within the domain of artificial intelligence. It has garnered attention for its attributes, including straightforward parallel processing-which results in significantly faster processing speeds compared to conventional GBM implementations-and exceptional predictive performance, making it a prominent contender among AI methodologies [42–44].

According to Tarwidi et al. [45], the core principle underlying the XGBoost algorithm is to minimize the following objective function, comprising both the loss function and regularization terms [45].

$$L^t = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t-1)}) + \Omega(f_t) \quad (2)$$

In this context, L , denotes the loss function capturing the disparity between observed data y_i and predicted data $\hat{y}_i$, where f_t represents the model t -three, and t denotes the iteration index during the optimization process. The specifics of the regularization term $\Omega(f)$ can be described as follows [45]:

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2 \quad (3)$$

Here, T denotes the total count of tree leaves, γ and λ represents penalty coefficients, and w is a vector encompassing the score of each leaf.

The training process yields the parameter f in Eq. (3), however before the training process starts, several hyperparameters including T , γ , and w need to be established. Many studies discovered that the accuracy of the model is significantly impacted by hyperparameter values. Consequently, it is necessary to find the ideal hyperparameters, which can be done by hyperparameter tuning. A selection of the XGBoost algorithm's most often tuned hyperparameters is displayed in Table 1.

3. Methodology

Our research endeavours to achieve precise electricity load forecasting by leveraging weather data and a signal decomposition technique applied to the electricity load data while using advanced boosting algorithms for modelling, notably XGBoost, AdaBoost and MLP, recognized for their strong predictive capabilities. Initial steps involve feature extraction on the electricity load data through spectrum analysis to identify primary load frequencies and consumer behaviour. Subsequently, focusing on the most influential frequency, we conduct signal decomposition using the Fourier transform on the load data to refine our predictive modelling approach. Subsequently, we conduct

Table 1
Commonly tuned XGBoost hyperparameters.

Hyperparameter	Type	Range	Default	Description
colsample_bytree	float	(0,1]	1	The percentage of features that will be used to build every tree
subsample	float	(0,1]	1	The percentage of each tree's data that will be sampled
learning_rate	float	[0,1]	0.3	Step size at each optimization of the objective function iteration
max_depth	int	[0, ∞)	6	Each tree's maximum depth; increasing this value complicates the model and increases the likelihood of overfitting.
n_estimators	int	[0, ∞)	100	The greatest quantity of gradient-boosted trees
min_child_weight	int	[0, ∞)	1	Required minimum instance weight for a child node
alpha	float	[0, ∞)	0	L1 regularization of weights. By raising this amount, the model will be conservatively adjusted.
lambda	float	[0, ∞)	1	L2 regularization of weights. By raising this amount, the model will be conservatively adjusted.
gamma	float	[0, ∞)	0	A tree leaf node's minimum loss reduction required to establish a new partition

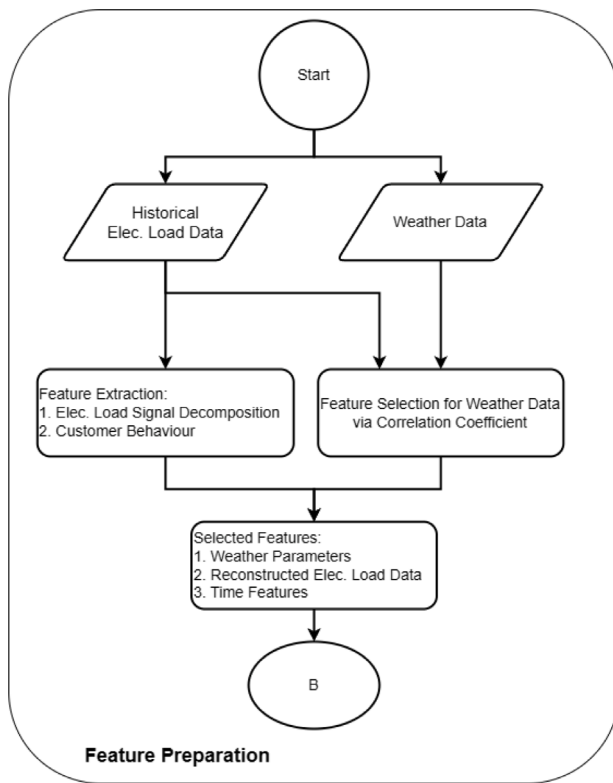


Fig. 1. Feature Preparation Flow Chart. Signal decomposition is conducted to reconstruct electricity load data to use it as a feature. Weather parameters with high CC values are also used as features.

feature selection from the ERA5 weather dataset by assessing the CC values between weather parameters and electricity load data. High CC values guide the selection of pertinent weather variables as features. The reconstructed electricity load data along with the time features and the high correlation ERA5 weather data are then utilized as inputs for training machine learning models.

In the next step, the machine learning model undergoes optimization. The selected electricity load and weather features are partitioned into training and testing datasets. The training data is employed to effectively train the model for accurate load forecasting, while the testing data assesses the model's predictive capacity. Our approach employs two prominent machine learning models, XGBoost and AdaBoost, tailored to excel in solving supervised learning problems, specifically

regression for electricity load forecasting. We aim to maximize these model's performance. In cases where the achieved performance falls below anticipated levels, we intend to refine the models through hyperparameter tuning using grid search, to enhance their predictive capabilities.

3.1. Feature preparation

At this stage, we pre-process the data that will serve as input features for the machine learning model. We acquire and prepare electricity load and weather data for feature extraction and selection. In the case of electricity load data, we apply spectrum analysis to analyse consumer behaviour and to extract the most significant signal frequencies. Subsequently, we select key frequencies and conduct signal decomposition using the Fourier Transform method to reconstruct the load data. From the spectrum analysis, the time feature and the reconstructed load data are then utilized as input features for the machine learning model. We determine the weather features to incorporate into the machine learning model by evaluating their CC values. Specifically, we choose weather parameters exhibiting a strong correlation with the electricity load data based on their high CC values with the electricity load data. The prepared electricity load, time and weather data are subsequently employed as features for the machine-learning models. The procedure of this stage is illustrated in Fig. 1

Additionally, this paper conducts spatial correlation analyses in two study areas, namely Bali and the Jakarta-Banten area, to investigate the influence of weather patterns in these regions. Two distinct test scenarios are employed: the first scenario entails identifying the point with the highest CC as the representative weather feature point, while the second scenario involves calculating the average CC across all points within each area. These analyses and their implications will be thoroughly discussed in Section 4.

3.2. Machine learning optimization

During this stage, our focus is on optimizing the machine learning model to ensure precise forecasting outcomes. Initially, the prepared data is segregated into training and testing datasets. The machine learning model is configured to align with the training data to enhance accuracy. Subsequently, the model undergoes training using the provided dataset, followed by testing on the reserved testing data to predict and assess the forecasted results using CC [46], RMSE (Root Mean Squared Error) [47], and MAPE (Mean Absolute Percentage Error) [48] values. If the forecasting outcomes fall short of expectations, we initiate hyperparameter tuning using grid search to refine the model's forecasting accuracy. The procedures at this stage are illustrated in Fig. 2.

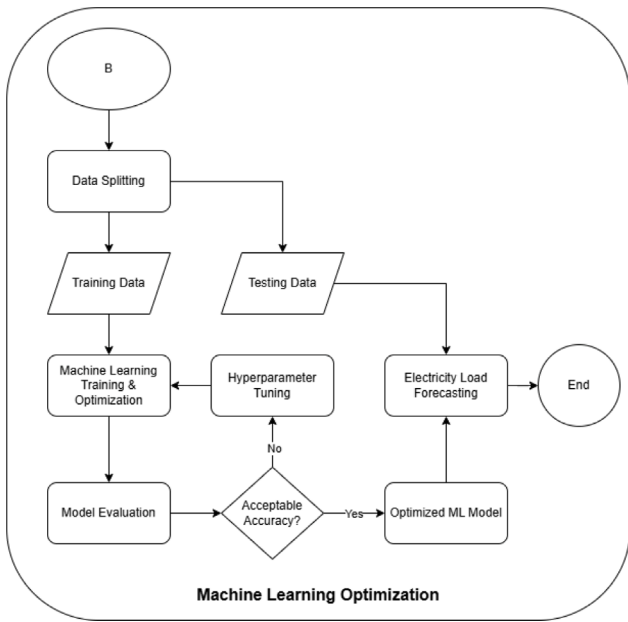


Fig. 2. Machine Learning Optimization Flow Chart. The prepared dataset is used for training the machine learning model. The model is optimized to achieve the best performance.

Table 2

Optimized value of the XBoost hyperparameters.

Hyperparameter	Search range	Optimal value
colsample_bytree	[0.6, 0.8, 1.0]	0.8
learning_rate	[0.01, 0.1, 0.3]	0.01
max_depth	[3, 5, 7]	5
n_estimators	[100, 200, 300]	300
subsample	[0.6, 0.8, 1.0]	0.8

3.2.1. XGBoost hyperparameter tuning

Hyperparameter tuning is performed to obtain the optimal hyperparameter value so that the model provides better forecasting accuracy. In this study, hyperparameter tuning is performed using the grid search method. The grid search method performs hyperparameter tuning by creating grids. To create grids, all possible pairings of the search interval's hyperparameters are provided during the grid search procedure. After that, the objective function in Eq. (2) is used to evaluate the error that each grid produces using the training dataset. The hyperparameters with the lowest error are the ideal ones. In this study, we perform hyperparameter optimization focusing on five important hyperparameters shown in Table 1, namely `colsample_bytree`, `learning_rate`, `max_depth`, `n_estimators` and `subsample`. After hyperparameter tuning using the grid search method by applying 5-fold-cross-validation, the optimal values obtained for the hyperparameters `colsample_bytree`, `learning_rate`, `max_depth`, `n_estimators` and `subsample` are 0.8, 0.01, 5, 300 and 0.8, respectively. The search ranges used for each hyperparameter and their optimal values can be seen in Table 2.

3.3. Electricity load data

This study focuses on analysing electricity load forecasting within two distinct regions characterized by unique electricity consumption patterns. Bali represents a relatively secluded region, while Jakarta-Banten boasts high electricity consumption and a diverse consumer base encompassing both households and industries. Our analysis centres on electricity load data extracted from the 'new normal' phase, denoting a period characterized by the resumption of typical routines alongside newly adopted habits. This dataset consists of 30 min interval

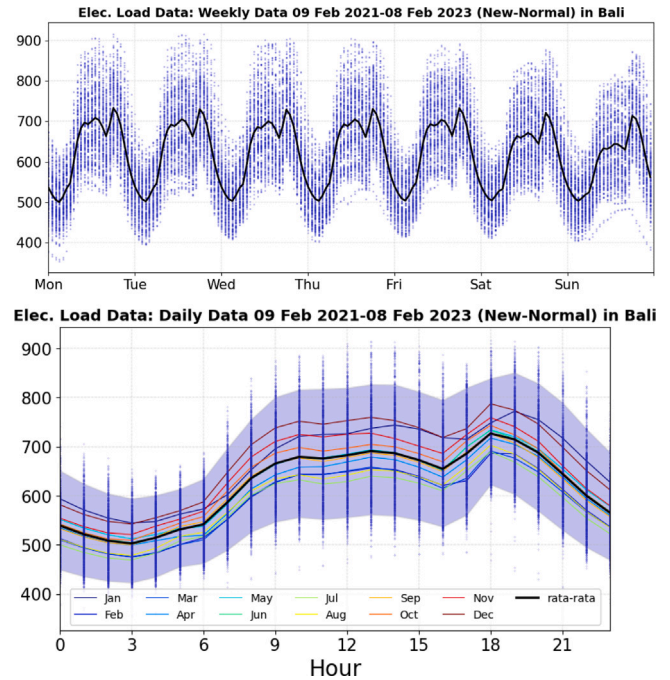


Fig. 3. Average weekly (top) and daily (bottom) variations in electricity load data observed in Bali throughout the new normal period (2021-02-09 to 2023-02-08).

data, spanning from February 9, 2021, to February 8, 2023. It offers a comprehensive view of electricity consumption patterns over this transitional period.

In Fig. 3, we present Bali's average weekly and daily electricity load data. The weekly electricity load data at the top of Fig. 3 demonstrates minimal differences in electricity usage, despite a decrease observed during weekends. Conversely, the daily electricity load data at the bottom of Fig. 3 reveals that the lowest average electricity consumption occurs at 3 o'clock, while the highest average usage peaks at 18 o'clock.

Fig. 4 presents a visualization of the electricity load in Bali, with grey bars indicating national holidays. The plot illustrates a consistent electricity load range of 300–1000 megawatts during the new-normal period, displaying a gradual increase in usage over time. Moreover, a notable trend emerges wherein electricity usage typically decreases during national holidays. We conducted an extensive analysis of the electricity load data by transforming it into the spectrum domain, revealing influential frequencies on the electricity load data, as depicted in Fig. 5. The analysis distinctly highlights the prominence of daily frequencies in influencing load patterns.

Fig. 6 presents Jakarta-Banten's average weekly and daily electricity load, offering valuable insights into its consumption patterns. In contrast to Bali, the weekly electricity load data at the top of Fig. 6 illustrates a significant difference in electricity usage between weekdays and weekends in Jakarta-Banten. Weekends exhibit a substantial decrease in electricity usage, coinciding with reduced business activities. Furthermore, the daily electricity load data at the bottom of Fig. 6 highlights that the lowest average electricity consumption occurs at 4 o'clock, while the highest average usage is observed at 14 o'clock.

Fig. 7 presents the electricity load plot for Jakarta-Banten, where the grey bars denote national holidays. A stark contrast to Bali is evident in Jakarta-Banten's electricity load, ranging between 4000–12000 megawatts owing to a larger consumer base in this region. Notably, the plot illustrates a distinct decline in electricity usage during national holidays. We conducted a comprehensive analysis of Jakarta-Banten's electricity load by transforming it into the spectrum domain as can be seen in Fig. 8 to identify influential frequencies shaping the load patterns. Similar to Bali, our analysis highlights the prominence of daily

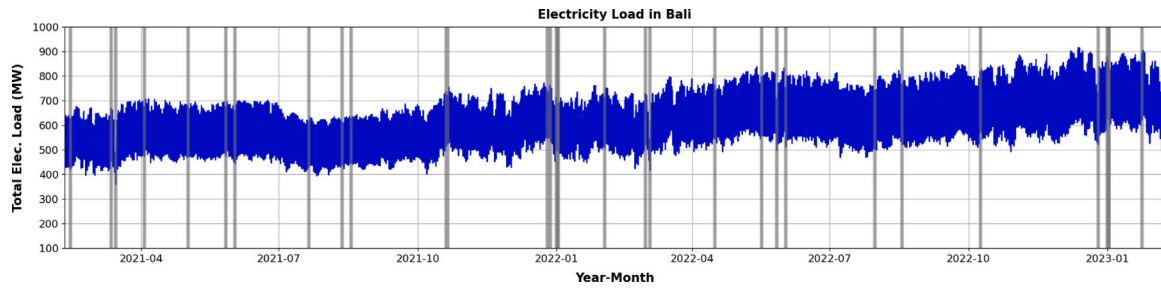


Fig. 4. Electricity load in Bali during the new-normal period (2021-02-09 to 2023-02-08). National holidays are marked with grey bars. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

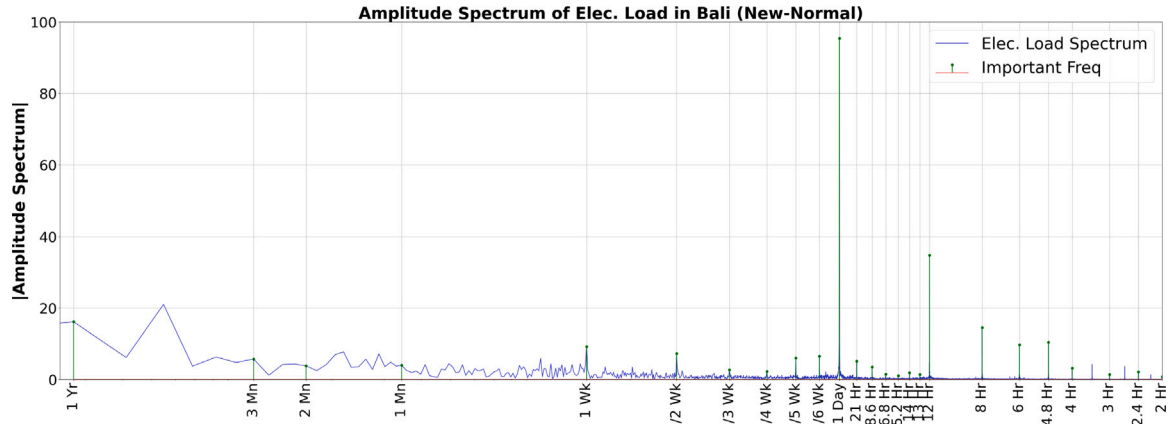


Fig. 5. Plot of the amplitude spectrum of the electricity load in Bali during the new-normal period (2021-02-09 to 2023-02-08).

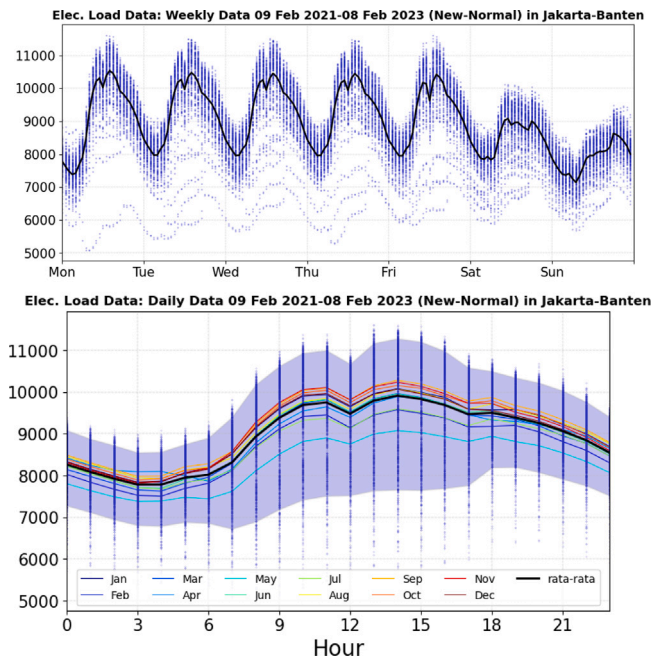


Fig. 6. Average weekly (top) and daily (bottom) variations in electricity load data observed in the Jakarta-Banten region throughout the new normal period (2021-02-09 to 2023-02-08).

frequencies, exhibiting the highest values, indicating their substantial impact on Jakarta-Banten's electricity load dynamics.

Analysis of electricity load data in both Bali and the Jakarta-Banten area reveals a consistent decrease in load during national holidays.

Recognizing the significance of this trend for accurate forecasting, we integrate holiday information as a time feature into our machine-learning model. Specifically, we categorize holiday information into four segments: regular holidays, Eid holidays, New Year holidays, and Nyepi holidays (exclusive to Bali). Furthermore, upon analysing the electric load spectrum in both regions, it was observed that the highest frequency or the most influential component in the electric load data occurs on a daily, hourly, and weekly basis. Recognizing the significance of this observation for precise forecasting, we have incorporated additional time features into our machine learning model inputs, including day of the week, day of the year, and specific day and hour timestamps. Historical electricity load data is a crucial factor in electricity load forecasting. Therefore, we aim to determine whether incorporating two weeks of historical electricity load data as a feature can enhance prediction accuracy. Additionally, leveraging crucial frequencies identified through spectrum analysis, we applied signal decomposition to reconstruct the electricity load data, utilizing these reconstructed load data as a feature to enhance the model's performance.

3.4. Weather data

In this research, the input features for the machine learning model comprise weather data obtained from the European Center for Medium-Range Weather Forecast (ECMWF) reanalysis, sourced from the ERA5 model [49]. The weather data is provided in hourly intervals. The considered weather parameters used as features for the machine learning model encompass temperature, solar radiation, wind speed, rainfall rate, pressure, and relative humidity.

To determine the relationship between weather parameters and electricity load, we used scatter plots and performed both linear and nonlinear regression analyses. Then, to determine the most appropriate weather parameters for integration as features within the machine

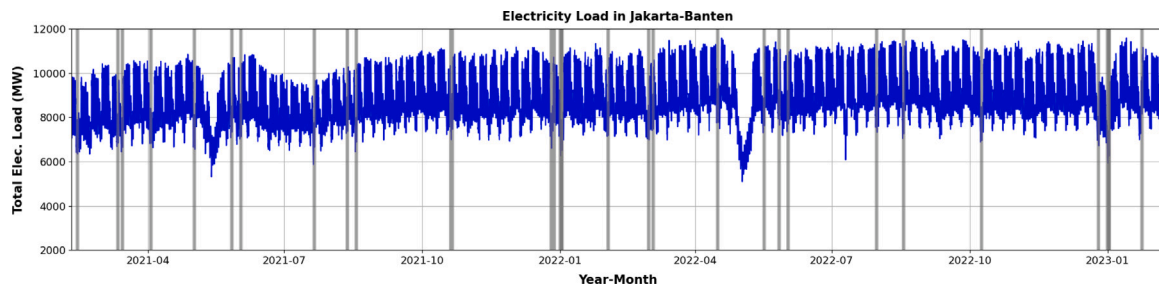


Fig. 7. Electricity load in Jakarta-Banten during the new-normal period (2021-02-09 to 2023-02-08). National holidays are marked with grey bars. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

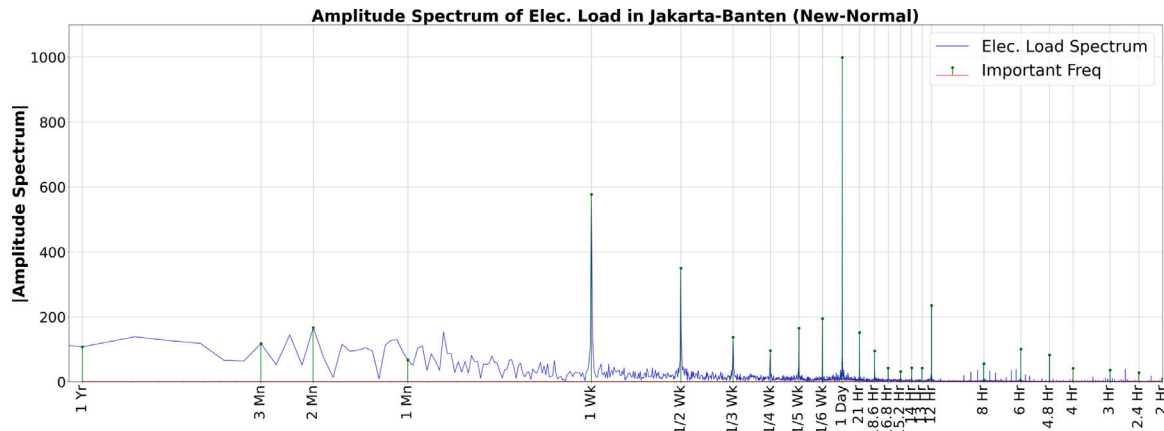


Fig. 8. Plot of the amplitude spectrum of the electricity load in Jakarta-Banten during the new-normal period (2021-02-09 to 2023-02-08).

learning model, we analyse the relationship between these parameters and electricity load data by computing their correlation values using the CC. This metric assists in gauging the degree of association between two variables, particularly in assessing the trend exhibited by these datasets.

3.4.1. Weather data of the Bali area

Fig. 11 depicts a comparative plot displaying the electricity load data in Bali throughout 2021 compared with weather parameters such as temperature, solar radiation, and wind speed. In Fig. 12, a similar comparison is showcased between the electricity load data and additional weather parameters, specifically rain rate, pressure, and relative humidity. Within both figures, the electricity load is graphically represented by a blue line, situated on the right-hand y-axis, while the respective weather parameters are denoted by red lines placed on the left-hand y-axis. A notable correspondence in trends is observed between temperature, solar radiation, and the electricity load in Bali, suggesting a high positive correlation between these parameters. Observing the correlation depicted at the bottom of Fig. 11 between wind speed and electricity load, a positive correlation is evident. However, it is noteworthy that the correlation between these two parameters is not particularly strong. In the upper section of Fig. 12, a low positive correlation between rain rate and electricity load is apparent. Moving to the midsection of Fig. 12, pressure exhibits a negative correlation with electricity load with a relatively weak correlation. Meanwhile, in the lower part of Fig. 12, the correlation between relative humidity and electricity load is stronger than that of pressure with electricity load.

Next, we performed linear and nonlinear regression analyses and visualized them using scatter plots to understand the relationship between weather parameters and electricity load. Figs. 9 and 10 present scatter plots along with the results of both linear and nonlinear regression analyses for each weather parameter in relation to electricity load. The findings indicate that nonlinear regression consistently outperformed linear regression across all weather parameters. The RMSE

Table 3

The results of Linear Regression (LR) and Nonlinear Regression (NLR) between weather parameters and electricity load in Bali. The NLR produced lower RMSE values. This finding suggests a nonlinear relationship between each weather parameter and electricity load.

Weather parameter	LR	NLR
	RMSE	RMSE
2 m Temperature	102.59	102.54
Net Solar Radiation	108.96	108.61
Wind Speed	118.55	118.31
Rainfall Rate	118.37	118.30
Pressure	118.33	118.23
Relative Humidity	111.94	110.25

values for both linear and nonlinear regression between weather parameters and electricity load are presented in [Table 3](#). Each weather parameter exhibits a lower RMSE in nonlinear regression, indicating that the relationship between weather parameters and electricity load is inherently nonlinear (see [Figs. 9](#) and [10](#)).

We present the CC values for each weather parameter in [Figs. 11 and 12](#). The CC values of person and spearman are outlined in [Table 4](#). This table demonstrates that the 2 m temperature boasts the highest CC among the weather parameters, with a CC value of 0.548, closely followed by solar radiation at 0.422. The CC values for additional weather parameters, specifically wind speed, rain rate, and pressure, are relatively small, with CC values of 0.028, 0.130, and -0.236 , respectively. However, in the case of humidity, the CC value stands higher compared to these three weather parameters, registering at -0.343 . Upon examination of the preliminary data, it is evident that temperature and solar radiation exhibit significant correlations with the electricity load data, with a CC value of 0.548 and 0.422, respectively, making them suitable candidates for feature selection in machine learning models for the Bali case.

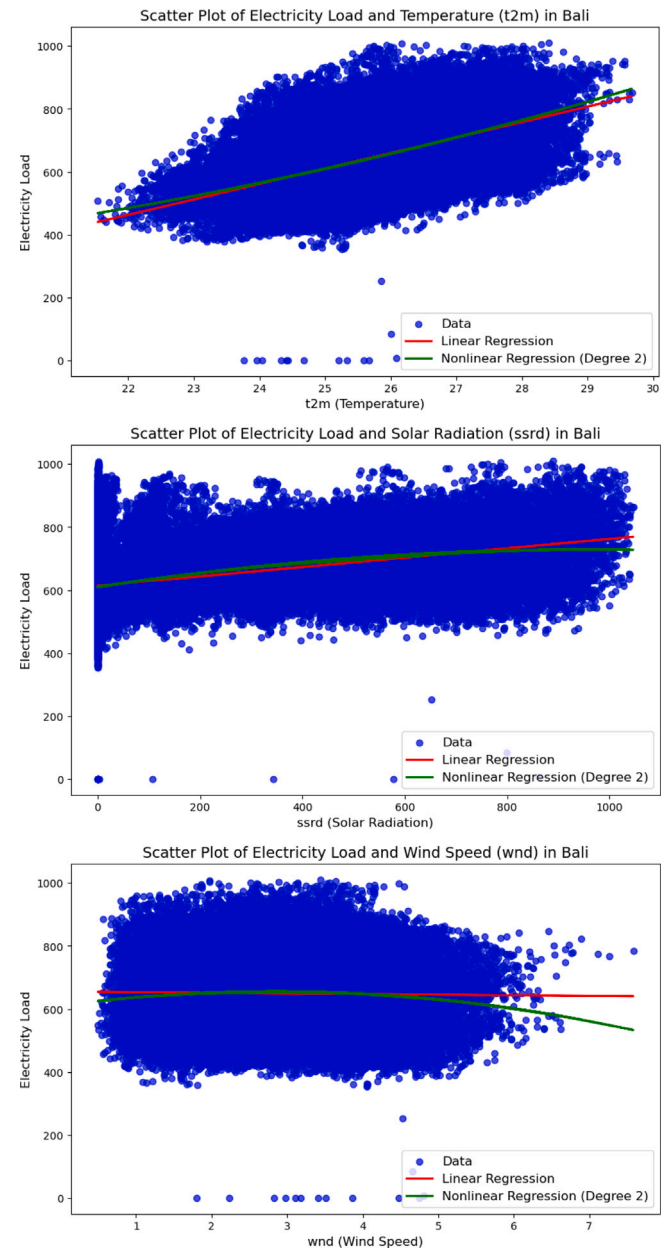


Fig. 9. Scatter plots and the results of linear and nonlinear regression for temperature (top), solar radiation (middle), and wind speed (bottom) in relation to electricity load in Bali. All weather parameters exhibit a nonlinear relationship with electricity load.

Weather parameter	CC	
	Pearson	Spearman
2 m Temperature	0.548	0.573
Net Solar Radiation	0.422	0.443
Wind Speed	0.028	0.030
Rainfall Rate	0.130	0.168
Pressure	−0.236	−0.225
Relative Humidity	−0.343	−0.399

3.4.2. Weather data of the Jakarta-Banten Area

Similar to the Bali case, we performed both linear and nonlinear regression analyses and visualized the results using scatter plots to examine the relationship between weather parameters and electricity

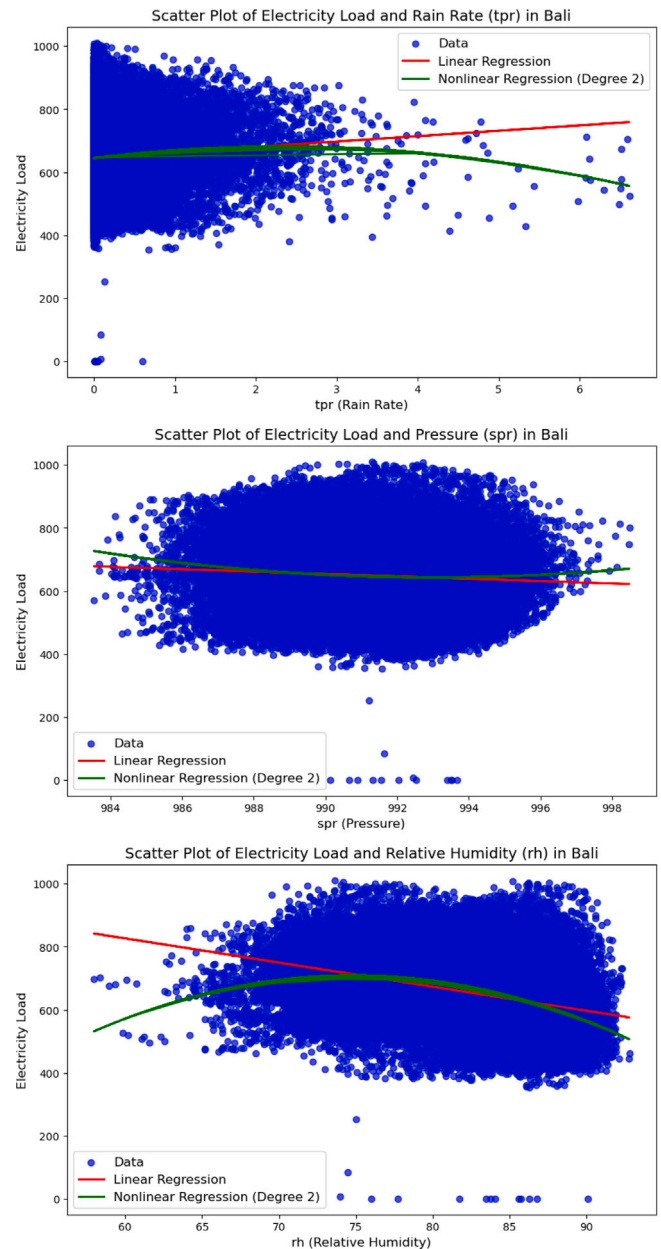


Fig. 10. Scatter plots and the results of linear and nonlinear regression for rain rate (top), pressure (middle), and relative humidity (bottom) in relation to electricity load in Bali. All weather parameters exhibit a nonlinear relationship with electricity load.

load. Figs. 13 and 14 illustrate the scatter plots alongside the regression outcomes for each weather parameter. The results consistently demonstrate that nonlinear regression performs better than linear regression across all weather parameters. Table 5 presents the RMSE values for both regression types, showing that nonlinear regression consistently yields lower RMSE values, reinforcing the inherently nonlinear relationship between weather parameters and electricity load.

Like in the case of Bali, our study delved into examining the CC values of weather parameters in the Jakarta-Banten area, aiming to ascertain the most suitable features for integration into the machine learning model. Fig. 15 presents a comparison plot showing the electricity load data in the Jakarta-Banten area throughout 2021 in comparison with essential weather parameters, including temperature, solar radiation, and wind speed. Similarly, Fig. 16 showcases a comparison between the electricity load data and additional weather parameters—rain rate, pressure, and relative humidity. Analogous to Bali, observable

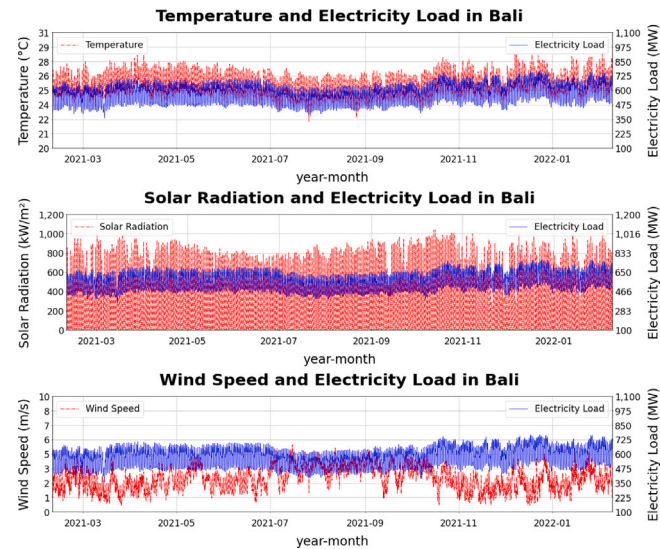


Fig. 11. Plot depicting the electricity load in comparison with various weather parameters in Bali for the year 2019: (Upper) temperature, (Middle) solar radiation, and (Lower) wind speed. The electric load measurements are shown on the right y-axis, while the weather parameter measurements are presented on the left y-axis.

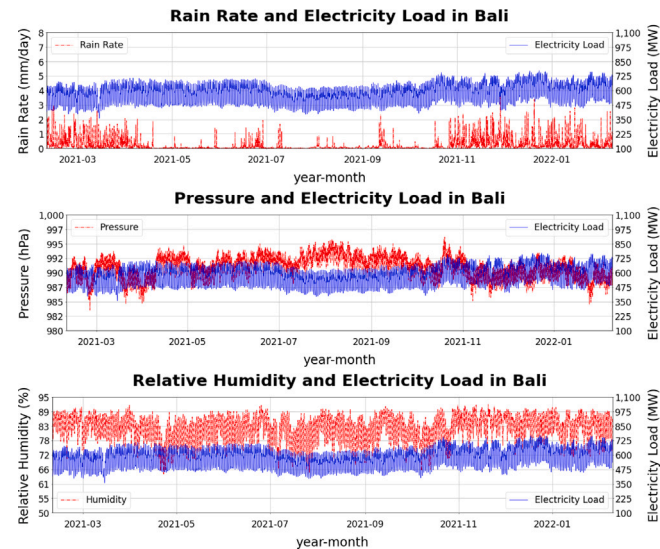


Fig. 12. Similarly to Fig. 11, other weather parameters are demonstrated, specifically: (Upper) rainfall rate, (Middle) pressure, and (Lower) relative humidity.

Weather parameter	LR	NLR
	RMSE	RMSE
2 m Temperature	1080.01	1077.34
Net Solar Radiation	1094.04	1088.50
Wind Speed	1199.78	1197.14
Rainfall Rate	1208.99	1208.99
Pressure	1210.19	1204.05
Relative Humidity	1057.97	1039.47

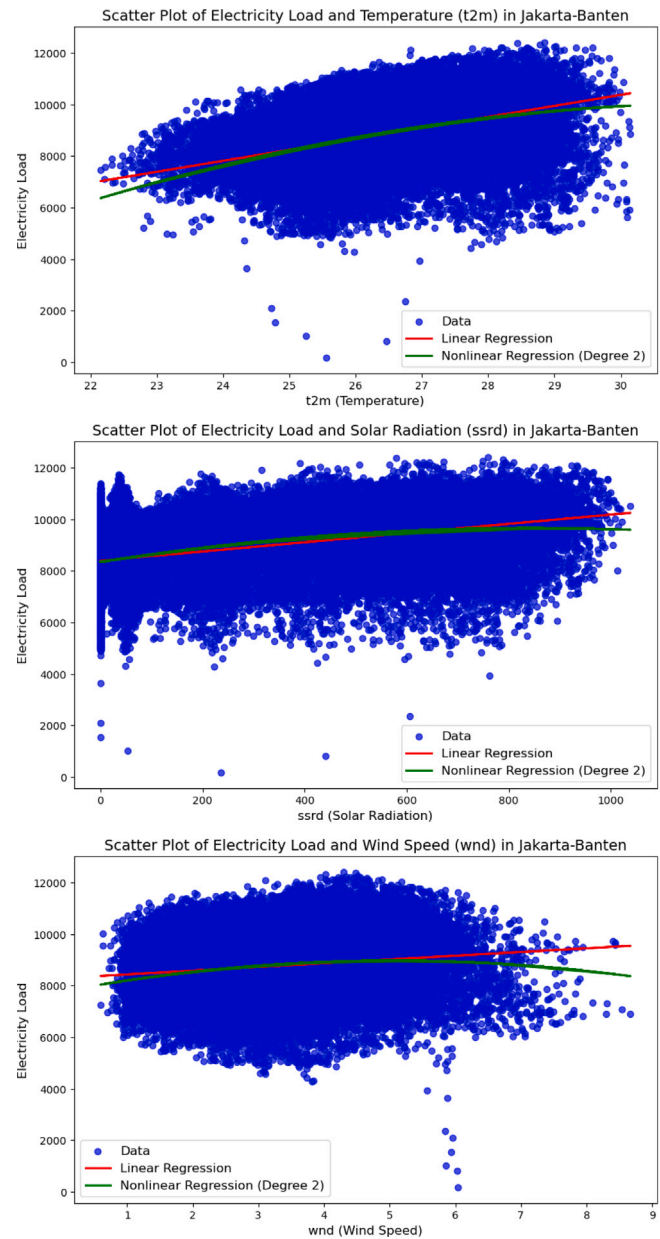


Fig. 13. Scatter plots and the results of linear and nonlinear regression for temperature (top), solar radiation (middle), and wind speed (bottom) in relation to electricity load in Jakarta-Banten. All weather parameters exhibit a nonlinear relationship with electricity load.

trends in the Jakarta-Banten area indicate noteworthy correlations between temperature, solar radiation, and electricity load, suggesting a high positive correlation among these variables. Similar to the scenario observed in Bali, the correlation analysis in the Jakarta-Banten area revealed a weak positive correlation between wind speed and electricity load. In the upper part of Fig. 16, a modest positive correlation between rain rate and electricity load can be observed. Transitioning to the middle part of Fig. 16, pressure exhibits a low negative correlation with electricity load. Moving to the lower part of Fig. 16, it shows a strong negative correlation between relative humidity and electricity load.

The CC values for each weather parameter, as demonstrated in Figs. 15 and 16, are presented in Table 6 along with the addition of spearman correlation values. The table reveals that the 2 m temperature and solar radiation demonstrate a high CC, with a CC value of 0.518 and 0.456, respectively. CC values for additional weather parameters

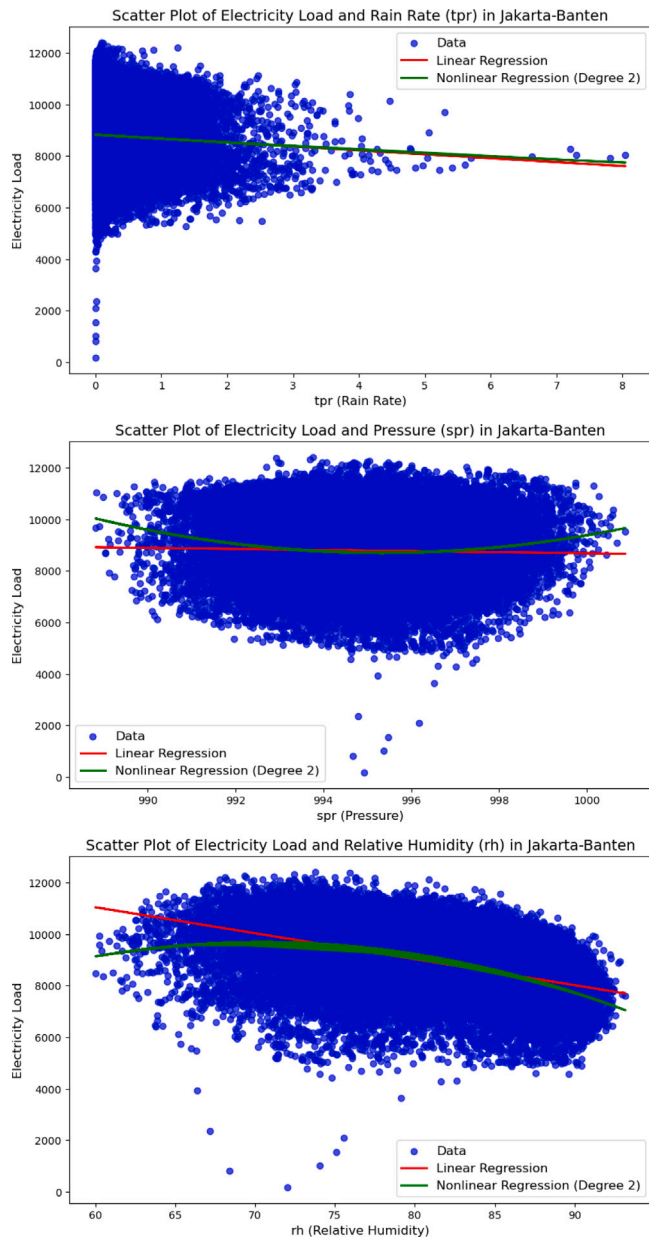


Fig. 14. Scatter plots and the results of linear and nonlinear regression for rain rate (top), pressure (middle), and relative humidity (bottom) in relation to electricity load in Jakarta-Banten. All weather parameters exhibit a nonlinear relationship with electricity load.

Table 6

Table of Correlation Coefficient (CC) between electricity load and various weather parameters in Jakarta-Banten.

Weather parameter	CC	
	Pearson	Spearman
2 m Temperature	0.518	0.546
Net Solar Radiation	0.456	0.419
Wind Speed	0.103	0.106
Rainfall Rate	0.012	0.035
Pressure	−0.119	−0.109
Relative Humidity	−0.510	−0.538

– specifically wind speed, rain rate, and pressure – indicate relatively small CC values of 0.103, 0.012, and −0.119, respectively. However, in the Jakarta-Banten area, relative humidity displays a high negative CC value, recorded at −0.510. Upon this exploratory data, it is apparent

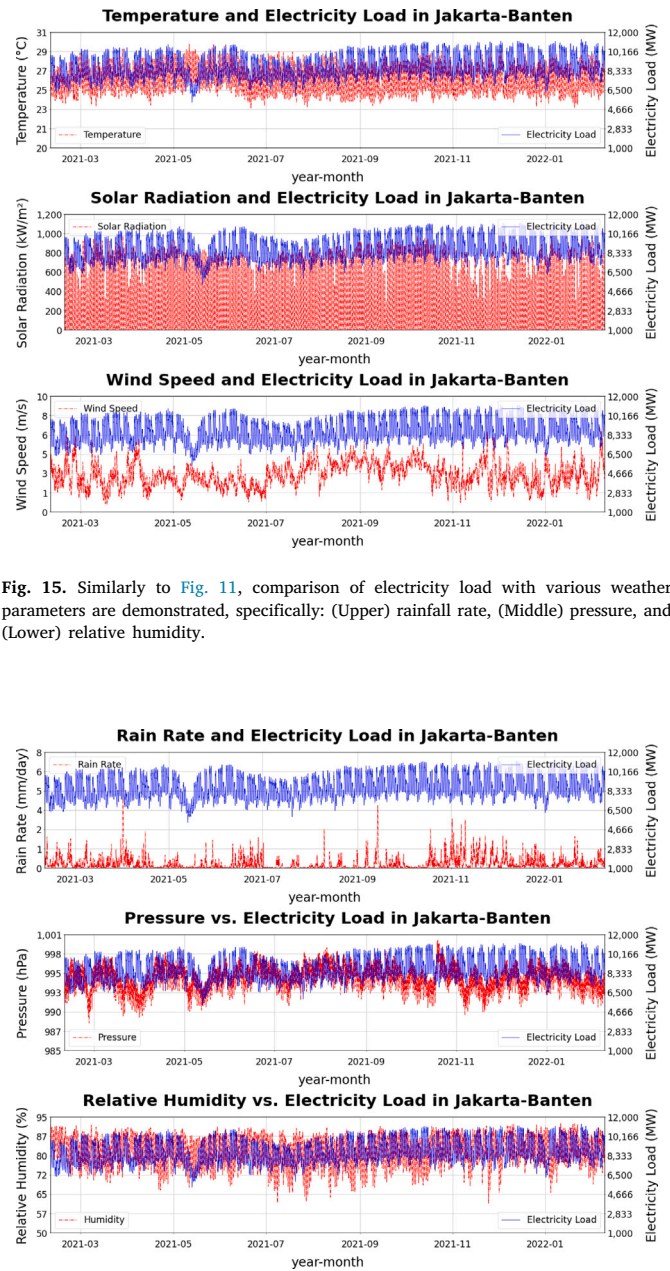


Fig. 15. Similarly to Fig. 11, comparison of electricity load with various weather parameters are demonstrated, specifically: (Upper) rainfall rate, (Middle) pressure, and (Lower) relative humidity.

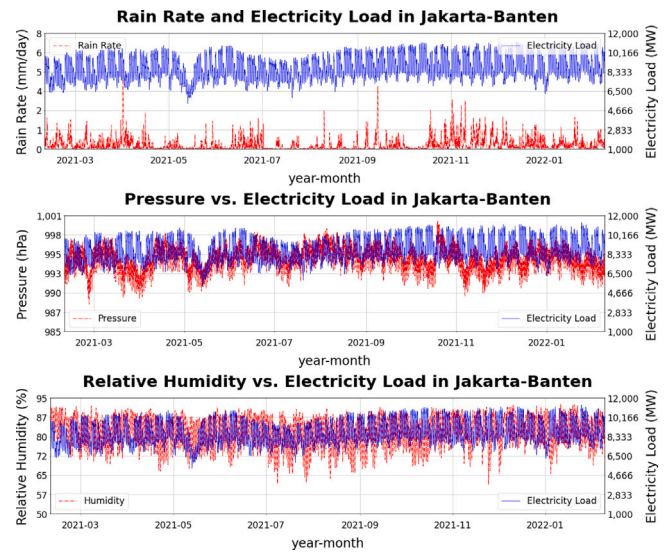


Fig. 16. Similarly to Fig. 15, other weather parameters are demonstrated, specifically: (Upper) rainfall rate, (Middle) pressure, and (Lower) relative humidity.

that temperature, solar radiation, and relative humidity are viable features for machine learning models in the Jakarta-Banten area. Our selection criteria for weather parameters in this case prioritize those demonstrating a positive correlation. Specifically, we focus on the 2 m Temperature, exhibiting a CC value of 0.518, and Net Solar Radiation, which shows a CC value of 0.456.

4. Results and discussion

In the previous section, we conducted a thorough examination of electricity load and weather data. Our analysis revealed a notable reduction in electricity load coinciding with holiday periods. Additionally, spectrum analysis unveiled daily, hourly, and weekly frequency components as exerting the greatest influence on electricity load dynamics. To enhance the input features of our machine learning model in light of these findings, we integrated a comprehensive time-feature

Table 7
XGBoost 14 days forecasting table results - Case Bali.

Model	Technique used for weather input values	Signal decomposition	Performance		
		Number of main frequency components	CC	RMSE (MW)	MAPE (%)
XGBoost	Avg	0	0.97	37.10	3.96
		5	0.97	35.49	3.82
		10	0.97	36.63	3.90
		15	0.97	36.83	3.91
		24	0.97	36.55	3.88
	Hi-CC	0	0.96	52.67	6.45
		5	0.96	41.19	4.79
		10	0.96	42.55	4.87
		15	0.96	42.69	4.87
		24	0.96	42.79	4.91

ensemble. This ensemble comprises holiday information, encompassing regular holidays, Eid holidays, New Year holidays, and Nyepi holidays (specifically for Bali), alongside temporal details such as day of the week, day of the year, day, and hour. Moreover, our analysis identified temperature and solar radiation as the weather parameters exhibiting the strongest correlation with electricity load. Consequently, we incorporated these two weather parameters as input features for our machine-learning model. In addition to the features mentioned above, historical electricity load data is a critical factor in electricity load forecasting.

This section presents the outcomes of our electricity load forecasting endeavours conducted across Bali and the Jakarta-Banten area. Our study encompasses diverse scenarios aimed at achieving optimal performance. Firstly, we delve into the comparison of weather input value selection techniques using the Spatial Correlation. This comparison involves two distinct methodologies: the first involves selecting a single weather point showcasing the highest correlation (referred to as ‘One Point with Highest Correlation Coefficient (Hi-CC)’), while the second technique utilizes weather input derived from averaging all spatial correlation within the respective area (‘Average Over Area (Avg)’).

Following this, we explore the impact of varying the number of main frequencies utilized in signal decomposition on our forecasting models. This step represents a key innovation in our research, where we employ the Fourier signal decomposition technique to enhance the accuracy of electricity load forecasting. The process involves a meticulous selection of key frequency components, identified through a thorough analysis of the spectral domain of the electrical load data. By focusing on the most significant frequencies – specifically the 5, 10, 15, and 24 highest amplitude signals – we isolate the most influential periodic patterns that directly contribute to variations in electricity demand. These selected frequencies are then used in the signal decomposition process, which allows us to filter out noise and non-essential fluctuations, thereby refining the input data fed into our forecasting models. This targeted decomposition not only improves the clarity of the signal but also enables the model to more accurately capture underlying trends and seasonal patterns in electricity consumption, ultimately leading to significantly enhanced predictive performance. Additionally, we will also investigate whether incorporating two weeks of historical electricity load data as a feature can enhance prediction accuracy.

4.1. Electricity load forecasting in Bali

4.1.1. Spatial correlation of the weather feature and the Optimal Main Frequency Component

Our initial investigation focused on assessing the sensitivity of selected weather locations. This exploration entailed conducting two distinct scenarios aimed at determining the optimal approach for utilizing weather features. The first scenario involved employing a single weather point exhibiting the highest correlation (Hi-CC) that can be seen in Fig. 17 where the highest point in the Bali area has a spatial

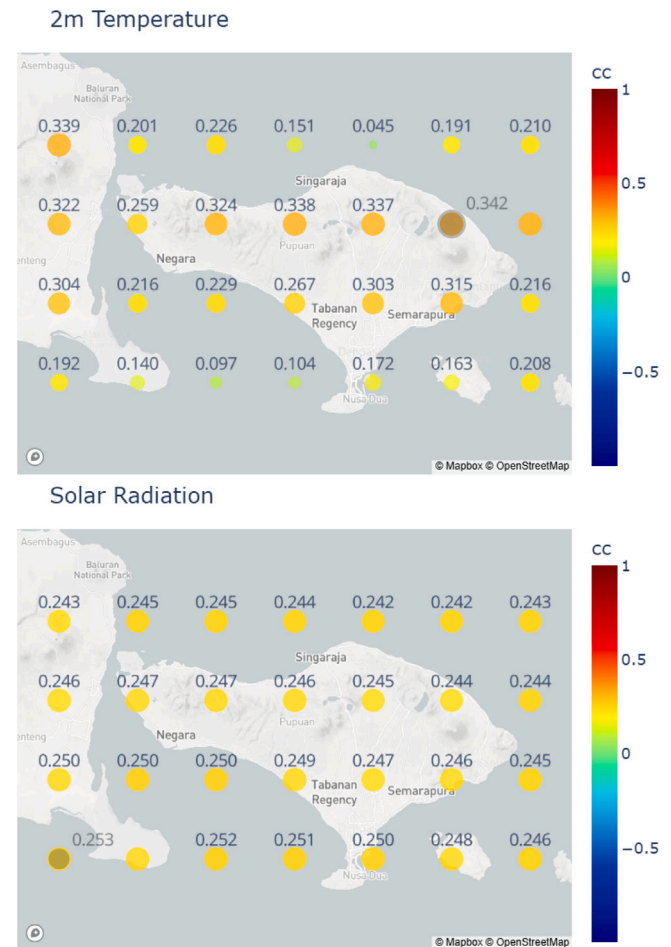


Fig. 17. Plot for the One Point with Highest Correlation Coefficient (Hi-CC) in Bali. The top figure shows the spatial correlation for 2 m temperature while the bottom figure shows the spatial correlation for solar radiation.

correlation of 0.342 for 2 m temperature and 0.253 for solar radiation, while the second scenario entailed using weather input calculated as the average of all spatial correlation within the respective area (Average Over Area (Avg)). In both scenarios, each model conducts a 14-day forecasting task. In configuring the XGBoost model, we specified the following hyperparameters: `colsample_bytree` set to 0.8, `learning_rate` set to 0.01, `max_depth` set to 5, `n_estimators` set to 300, and `subsample` set to 0.8. As for the AdaBoost model, the hyperparameters were set with the base estimator as a decision tree regressor and `n_estimators` as 50. The test outcomes for both scenarios are illustrated in Table 7 and Table 8.

Table 8
AdaBoost 14 days forecasting table results - Case Bali.

Model	Technique used for weather input values	Signal decomposition	Performance		
		Number of main frequency components	CC	RMSE (MW)	MAPE (%)
AdaBoost	Avg	0	0.96	36.10	3.83
		5	0.95	35.87	3.85
		10	0.96	36.30	3.76
		15	0.96	35.10	3.59
		24	0.96	35.15	3.66
	Hi-CC	0	0.93	46.77	5.58
		5	0.93	40.31	4.52
		10	0.94	40.27	4.44
		15	0.93	41.65	4.67
		24	0.94	41.41	4.64

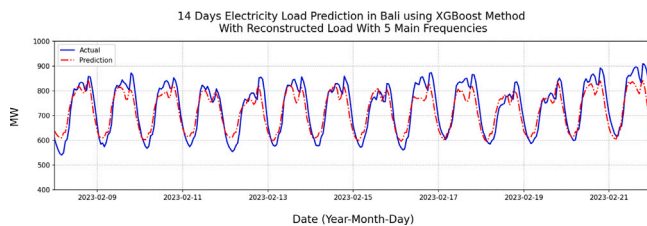


Fig. 18. Plot depicting the comparison between actual and predicted 14-day electricity load data in Bali using XGBoost with five main frequencies for signal decomposition.

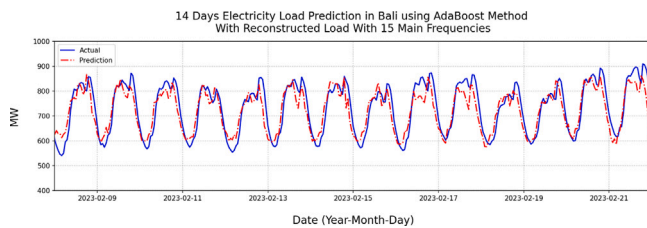


Fig. 19. Plot depicting the comparison between actual and predicted 14-day electricity load data in Bali using AdaBoost with fifteen main frequencies for signal decomposition.

After comparison, it was evident that using weather inputs through the ‘Average Over Area’ scenario yields superior results for both models. In the case of XGBoost, the optimal performance is achieved by a model comprising 5 main frequency components, exhibiting commendable CC, RMSE, and MAPE values of 0.97, 35.49, and 3.82, respectively. The plot regarding the comparison of the predicted and actual data using XGBoost with 5 main frequencies for signal decomposition can be seen in Fig. 18. Meanwhile, for AdaBoost, the most effective performance is demonstrated by a model comprising 15 main frequency components, showcasing noteworthy CC, RMSE, and MAPE values of 0.96, 35.10, and 3.59, respectively. The plot regarding the comparison of the predicted and actual data using AdaBoost with 15 main frequencies for signal decomposition can be seen in Fig. 19.

4.1.2. Multi Days Forecasting with the Optimal Main Frequency Component

In the subsequent phase, we conducted a comparison for each model using the optimal number of main frequencies to perform multi-day forecasting. In the case of XGBoost, the highest result was obtained from five main frequencies for signal decomposition, displaying notable CC, RMSE, and MAPE values of 0.97, 35.49, and 3.82. For AdaBoost, the model achieved optimal signal decomposition with 15 main frequencies, demonstrating substantial CC, RMSE, and MAPE values of 0.96, 35.10, and 3.59, respectively. Each model will undertake forecasting tasks for intervals of 1, 3, 7, and 14 days.

After conducting multi-day forecasting using both models, XGBoost demonstrated its optimal performance in 3-day forecasting by employing five main frequencies for signal decomposition. This model showcased R^2 , CC, RMSE, and MAPE values of 0.89, 0.98, 37.83, and 4.66, respectively. In AdaBoost, employing 15 main frequencies for signal decomposition yielded its peak performance in the context of 7-day forecasting, displaying noteworthy R^2 , CC, RMSE, and MAPE values of 0.90, 0.97, 32.00, and 3.84, respectively. The comprehensive outcomes of the XGBoost and AdaBoost forecasting experiments for multiple days are provided in Table 10. Analysis of the experimental outcomes indicates that, for the Bali case, overall, XGBoost demonstrates superior performance in electricity load forecasting compared to AdaBoost.

To further gauge the efficacy of XGBoost in electricity load forecasting, we conducted a comparative analysis between XGBoost and a deep learning algorithm, specifically the Multi-Layer Perceptron (MLP). For the MLP model, we configured the hyperparameters as follows: hidden_layer_sizes set to (125, 25, 5), activation function set to relu, solver set to adam, and n_estimators set to 500. Before the comparison, the Multi-Layer Perceptron (MLP) underwent experimentation akin to the preceding models. Notably, the optimal performance was observed in MLP utilizing five main frequencies for signal decomposition and adopting the ‘Average Over Area’ technique for selecting weather data input values. This result can be seen in Table 9.

Following the comparative analysis with MLP, XGBoost continues to exhibit superior performance, particularly excelling in 3-day forecasting with five main frequencies for signal decomposition, displaying notable R^2 , CC, RMSE, and MAPE values of 0.89, 0.98, 37.83, and 4.66, respectively. The MLP model demonstrates its peak performance in 7-day forecasting, using five main frequencies for signal decomposition, presenting competitive R^2 , CC, RMSE, and MAPE values of 0.90, 0.97, 33.69, and 3.86, respectively. The comprehensive forecasting results of both models are presented in Table 10 for detailed examination. Examination of the experimental results reveals that, in the case of Bali, XGBoost generally outperforms MLP in electricity load forecasting.

4.1.3. Two weeks historical load data as feature

In the next stage, we evaluate whether incorporating two weeks of historical electricity load data as a feature can improve prediction accuracy. The results of models that do not use this feature are compared with those that do, to assess the impact of including two weeks of historical load data on model performance. After incorporating two weeks of historical electricity load data as a feature, the overall forecasting accuracy decreased. This indicates that, in the case of Bali, the use of historical data did not enhance prediction accuracy. The results of this comparison are presented in Table 11.

Through a series of extensive experiments, it is evident that in the context of Bali’s electricity load forecasting, XGBoost with five main frequencies for signal decomposition outperforms other boosting

Table 9
MLP 14 days forecasting table results - Case Bali.

Model	Technique used for weather input values	Signal Decomposition	Performance		
		Number of main frequency components	CC	RMSE (MW)	MAPE (%)
MLP	Avg	0	0.97	37.55	3.98
		5	0.97	36.11	3.80
		10	0.96	38.33	4.10
		15	0.96	36.35	3.83
		24	0.95	35.40	3.93
	Hi-CC	0	0.96	39.76	4.23
		5	0.95	35.72	3.99
		10	0.95	38.17	4.09
		15	0.95	39.01	4.18
		24	0.96	36.84	3.95

Table 10
Table results of forecasting with various length (days) - Case Bali.

Model	Technique used for weather input values	Signal decomposition	Forecast length (Days)	Performance		
		Number of main frequency components		CC	RMSE	MAPE (%)
XGBoost	Avg	5	1	0.98	41.20	4.97
			3	0.98	37.83	4.66
			7	0.97	37.29	4.53
			14	0.97	35.49	3.82
AdaBoost	Avg	15	1	0.95	42.91	5.07
			3	0.94	39.47	4.62
			7	0.97	32.00	3.84
			14	0.96	35.10	3.59
MLP	Avg	5	1	0.96	44.94	5.29
			3	0.97	35.75	4.28
			7	0.97	33.69	3.86
			14	0.97	36.11	3.80

Table 11
Forecasting result in Bali with and without historical electricity load data as feature.

Model	Forecast length (Days)	Performance with 2-week Historical Load Data as Features							
		R ²	CC	RMSE	MAPE (%)	R ²	CC	RMSE	MAPE (%)
XGBoost	1	0.89	0.98	41.20	4.97	0.90	0.96	43.22	5.24
	3	0.89	0.98	37.83	4.53	0.90	0.97	39.30	4.84
	7	0.90	0.97	37.29	4.66	0.90	0.97	37.34	4.57
	14	0.90	0.97	35.49	3.82	0.90	0.97	36.74	3.92
AdaBoost	1	0.83	0.95	42.91	5.07	0.86	0.91	47.62	5.77
	3	0.90	0.94	39.47	4.62	0.90	0.94	42.44	5.07
	7	0.90	0.97	32.00	3.84	0.90	0.96	36.68	4.51
	14	0.90	0.96	35.10	3.59	0.90	0.96	36.85	3.82
MLP	1	0.90	0.96	44.94	5.29	0.86	0.91	50.13	5.79
	3	0.90	0.97	35.75	4.28	0.90	0.94	42.48	4.86
	7	0.90	0.97	33.69	3.86	0.90	0.94	38.67	4.35
	14	0.90	0.97	36.11	3.80	0.90	0.94	40.64	4.25

algorithms such as AdaBoost and deep learning models like Multi-Layer Perceptron (MLP), delivering the most accurate and reliable results. Additionally, the results showed that incorporating two weeks of historical electricity load data as a feature did not improve prediction accuracy.

4.2. Electricity load forecasting in Jakarta-Banten Area

4.2.1. Spatial correlation of the weather feature and the optimal main frequency component

Similar to our investigations in the Bali region, our analysis in the Jakarta-Banten area began by assessing the sensitivity of selected weather locations. This exploration encompassed two distinctive scenarios aimed at optimizing the utilization of weather features. The initial scenario involved employing a singular weather point displaying the highest correlation (Hi-CC) that can be seen in Fig. 20 where the highest point in the Jakarta-Banten area has a spatial correlation of 0.526 for 2 m Temperature. For the solar radiation, since the highest

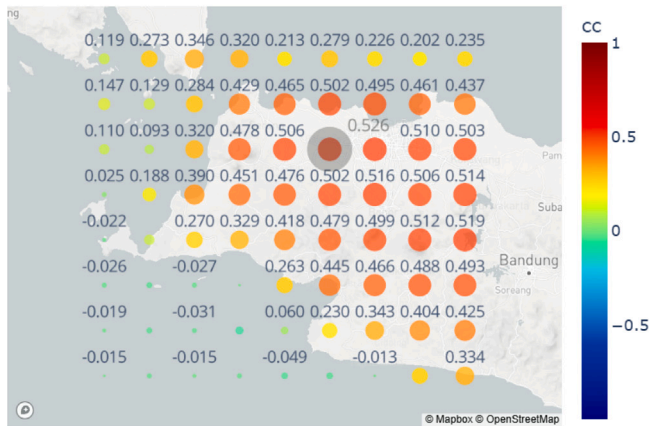
correlation is at the ocean and too far from the investigated area, we chose a correlation point close to the investigated area. The spatial correlation for the solar radiation is 0.386. The second scenario utilized weather input calculated as the average of all spatial correlation within the respective area (Avg). Each model conducted a 14-day forecasting task in both scenarios. For the XGBoost model, we utilized its default hyperparameter settings, which include `colsample_bytree` at 1, `learning_rate` at 0.3, `max_depth` at 6, `n_estimators` at 100, and `subsample` at 1. As for the AdaBoost model, we configured the hyperparameters identically to the setting used for the Bali case, with the `base_estimator` as a decision tree regressor and `n_estimators` set to 50. The results of the forecasting are summarized in Tables 12 and 13.

Upon comparison, it was evident that adopting the 'Average Over Area' scenario for weather inputs led to superior results for both XGBoost and AdaBoost models. Specifically, in the case of XGBoost, the most effective performance was observed in a model comprising five main frequency components, showcasing notable CC, RMSE, and MAPE values of 0.95, 497.99, and 4.49, respectively. The plot regarding

Table 12
XGBoost 14 days forecasting table results - Case Jakarta-Banten.

Model	Technique Used for Weather Input Values	Signal Decomposition	Performance		
		Number of main frequency components	CC	RMSE (MW)	MAPE (%)
XGBoost	Avg	0	0.94	632.07	6.02
		5	0.95	497.99	4.49
		10	0.94	538.88	4.89
		15	0.94	545.86	5.09
		24	0.94	546.52	5.04
	Hi-CC	0	0.90	707.30	6.84
		5	0.90	630.65	5.94
		10	0.89	673.00	6.38
		15	0.89	682.93	6.56
		24	0.89	681.35	6.53

2m Temperature



Solar Radiation

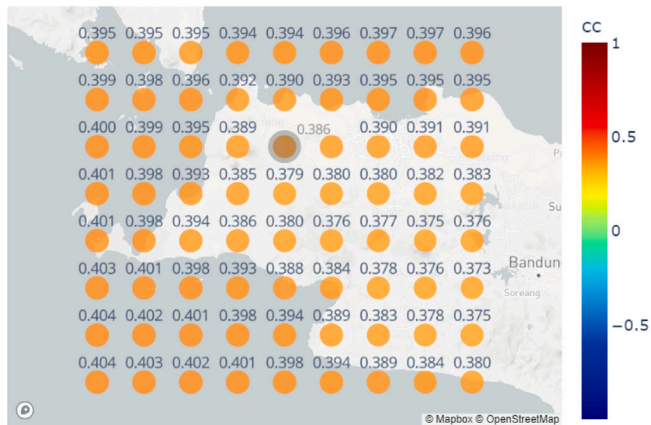


Fig. 20. Plot for the One Point with Highest Correlation Coefficient (Hi-CC) in Jakarta-Banten. The top figure shows the spatial correlation for 2 m temperature while the bottom figure shows the spatial correlation for solar radiation.

the comparison of the predicted and actual data using XGBoost with five main frequencies for signal decomposition can be seen in Fig. 21. On the other hand, for AdaBoost, the model's optimal performance was also achieved by employing five main frequencies for signal decomposition, displaying commendable CC, RMSE, and MAPE values of 0.95, 523.95, and 4.82, respectively. The plot regarding the comparison of the predicted and actual data using AdaBoost with five main frequencies for signal decomposition can be seen in Fig. 22.

4.2.2. Multi days forecasting with the optimal main frequency component

Following this, a comparative analysis for each model employing the optimal number of main frequencies was conducted to facilitate

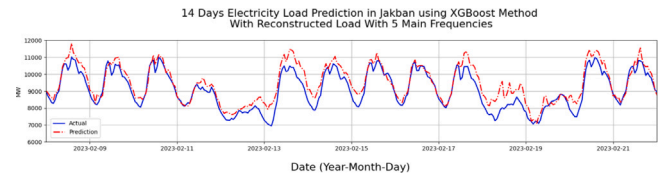


Fig. 21. Plot depicting the comparison between actual and predicted 14-day electricity load data in Jakarta-Banten using XGBoost with five main frequencies for signal decomposition.



Fig. 22. Plot depicting the comparison between actual and predicted 14-day electricity load data in Jakarta-Banten using AdaBoost with five main frequencies for signal decomposition.

multi-day forecasting. In the context of XGBoost, the most effective signal decomposition involved five main frequencies, presenting notable CC, RMSE, and MAPE values of 0.95, 497.99, and 4.49. For AdaBoost, the model also exhibited optimal performance with five main frequencies, demonstrating significant CC, RMSE, and MAPE values of 0.95, 523.95, and 4.82, respectively. Both models will undertake forecasting for intervals of 1, 3, 7, and 14 days.

After conducting multi-day forecasting using both models, XGBoost demonstrated its superior performance in 14-day forecasting by employing five main frequencies for signal decomposition, showcasing exceptional R^2 , CC, RMSE, and MAPE values of 0.90, 0.95, 497.99, and 4.49, respectively. In the AdaBoost model, employing five main frequencies for signal decomposition resulted in optimal performance in the context of 14-day forecasting, exhibiting noteworthy R^2 , CC, RMSE, and MAPE values of 0.90, 0.95, 523.95, and 4.82, respectively. The comprehensive results of these forecasting experiments for multiple days are tabulated in Table 15. Following a comprehensive series of experiments, it is evident that, akin to the Bali region, in the context of electricity load forecasting in the Jakarta-Banten area, XGBoost demonstrates superior performance compared to AdaBoost.

To further evaluate the efficacy of XGBoost in electricity load forecasting, the same as in the Bali case, we also conducted a comparative analysis between XGBoost and a deep learning algorithm, specifically the Multi-Layer Perceptron (MLP). For the MLP model, we maintained consistent hyperparameter settings as those used for the Bali case, specifying hidden_layer_sizes as (125, 25, 5), activation function as relu, solver as adam, and n_estimators as 500. Preceding the comparison, the

Table 13
AdaBoost 14 days forecasting table results - Case Jakarta-Banten.

Model	Technique Used for Weather Input Values	Signal Decomposition	Performance		
		Number of Main Frequency Components	CC	RMSE (MW)	MAPE (%)
AdaBoost	Avg	0	0.94	613.83	5.86
		5	0.95	523.95	4.82
		10	0.95	549.07	5.17
		15	0.94	569.63	5.35
		24	0.94	572.33	5.40
	Hi-CC	0	0.85	623.76	5.77
		5	0.84	627.14	5.78
		10	0.84	626.14	5.80
		15	0.83	637.90	5.88
		24	0.82	649.71	5.94

Table 14
MLP 14 days forecasting table results - Case Jakarta-Banten.

Model	Technique Used for Weather Input Values	Signal Decomposition	Performance		
		Number of Main Frequency Components	CC	RMSE (MW)	MAPE (%)
MLP	Avg	0	0.94	533.88	4.87
		5	0.92	511.88	4.11
		10	0.93	496.34	4.27
		15	0.92	544.64	4.75
		24	0.92	572.48	5.10
	Hi-CC	0	0.93	499.60	4.77
		5	0.91	508.27	4.60
		10	0.91	514.27	4.72
		15	0.91	531.06	4.92
		24	0.73	839.91	7.72

Table 15
Table results of forecasting with various length (days) - Case Jakarta-Banten.

Model	Technique used for weather input values	Signal decomposition	Forecast length (Days)	Performance		
		Number of main frequency components		CC	RMSE	MAPE (%)
XGBoost	Avg	5	1	0.94	407.74	3.73
			3	0.94	445.10	4.14
			7	0.94	551.43	5.15
			14	0.95	497.99	4.49
AdaBoost	Avg	5	1	0.76	620.98	5.39
			3	0.90	544.80	5.15
			7	0.95	527.87	4.98
			14	0.95	523.95	4.82
MLP	Avg	10	1	0.83	586.79	4.56
			3	0.86	514.46	4.32
			7	0.91	495.94	4.34
			14	0.93	496.34	4.27

Multi-Layer Perceptron (MLP) was subjected to experimental analysis, akin to the previous models. Notably, the optimal performance was observed in the MLP model employing ten main frequencies for signal decomposition and utilizing the ‘Average Over Area’ technique for selecting weather data input values. This result can be seen in [Table 14](#).

Upon detailed comparison between XGBoost and MLP, it is evident that both models exhibit optimal performance in 14-day forecasting. XGBoost yields R^2 , CC, RMSE, and MAPE results of 0.90, 0.95, 497.99, and 4.49, respectively, while MLP produces R^2 , CC, RMSE, and MAPE results of 0.89, 0.93, 496.34, and 4.27, respectively. Overall, the XGBoost demonstrates superior performance compared to MLP. The comprehensive forecasting results of both models are outlined in [Table 15](#) for detailed examination.

4.2.3. Two weeks historical load data as feature

In the next stage, we incorporated two weeks of historical electricity load data as a feature to assess potential improvements in prediction accuracy. For some parts, this addition enhances prediction accuracy. There was an improvement in the XGBoost model for 1-day forecasting.

Without the two weeks of historical data, the R^2 , CC, RMSE, and MAPE results were 0.90, 0.94, 407.74, and 3.73, respectively. After including the historical data, these metrics improved to 0.90, 0.95, 406.60, and 3.55. However, overall, the model performance decreased when an additional two weeks of historical load data was used as a feature. The results can be seen in [Table 16](#).

Through an extensive series of experiments, it is evident that XGBoost with five main frequencies for signal decomposition outperforms other boosting algorithms such as AdaBoost and deep learning models like Multi-Layer Perceptron (MLP) in the context of the Jakarta-Banten area, delivering the most accurate and reliable forecasting results. Additionally, the incorporation of two weeks of historical electricity load data did not improve the forecasting accuracy.

5. Conclusion

Machine learning serves as a valuable tool for electricity load forecasting, and the key to achieving accurate predictions lies in the careful selection of pertinent features. This involves a thorough exploration of both electricity load data and its associated factors, such

Table 16
Forecasting result in Jakarta-Banten with and without historical electricity load data as feature.

Model	Forecast length (Days)	Performance with 2-week historical load data as features							
		R ²	CC	RMSE	MAPE (%)	R ²	CC	RMSE	MAPE (%)
XGBoost	1	0.90	0.94	407.74	3.73	0.90	0.95	406.60	3.55
	3	0.90	0.94	445.10	4.14	0.90	0.95	445.21	4.20
	7	0.90	0.94	551.43	5.15	0.90	0.95	553.85	5.23
	14	0.90	0.95	497.99	4.49	0.90	0.94	560.67	5.10
AdaBoost	1	0.01	0.76	620.98	5.39	0.60	0.84	503.32	3.86
	3	0.83	0.90	544.80	5.15	0.87	0.92	528.65	4.87
	7	0.90	0.95	527.87	4.98	0.90	0.94	563.69	5.16
	14	0.90	0.95	523.95	4.82	0.90	0.94	552.64	5.08
MLP	1	0.54	0.83	586.79	4.56	0.63	0.86	572.43	4.45
	3	0.63	0.86	514.46	4.32	0.86	0.91	506.82	4.30
	7	0.86	0.91	495.94	4.34	0.86	0.91	503.72	4.61
	14	0.89	0.93	496.34	4.27	0.86	0.93	465.35	4.08

as weather parameters. To enhance the accuracy of short-term load forecasting (STLF) in power system operations, feature engineering methods must consider temporal dynamics and external meteorological factors. Furthermore, enhancing the accuracy of the forecasting system involves simplifying the electricity load data through signal decomposition techniques.

In this study, spectrum analysis is employed on the electricity load data to observe consumer behaviour and to identify and extract crucial frequencies for subsequent signal decomposition. Additionally, data exploration is conducted on weather data to discern its correlation with electricity load data, further contributing to the refinement of our forecasting model. This paper also explores a method for selecting weather sites based on spatial correlation analysis. XGBoost and AdaBoost algorithms were used for electricity load forecasting in two distinct locales, Bali and the Jakarta-Banten area, each characterized by unique traits. Additionally, Multi-Layer Perceptron (MLP) is employed for comparative analysis. Across both locations, XGBoost consistently outperforms AdaBoost and MLP, demonstrating superior forecasting capabilities. Notably, in Bali, the optimal performance is achieved by XGBoost employing 'Average Over Area (Avg)' for weather data input selection and utilizing five main frequencies in signal decomposition. This configuration yields the most accurate predictions for the next three days, exhibiting impressive values of 0.89 for R^2 , 0.98 for CC, 37.83 for RMSE, and 4.66 for MAPE. In the context of Jakarta-Banten, XGBoost attains superior performance, employing 'Average Over Area (Avg)' for selecting weather data inputs and employing five main frequencies for signal decomposition. This configuration yields outstanding predictive accuracy for a 14-day forecasting horizon, showcasing impressive values of 0.90 for R^2 , 0.95 for CC, 497.99 for RMSE, and 4.49 for MAPE. Additionally, the results indicate that using historical electricity load data as a feature did not improve prediction accuracy for either case.

Our findings reveal that the highest prediction accuracy is achieved by averaging the correlation coefficients across the study area, both in the context of Bali and Jakarta. Furthermore, incorporating signal decomposition using Fourier transformation with five main frequencies demonstrates a notable improvement in prediction accuracy. Specifically, in the Bali case, the addition of signal decomposition results in a 4.6% reduction in RMSE for 14-day prediction using the XGBoost model. Similarly, in the Jakarta-Banten area case, the RMSE for 14-day prediction using the XGBoost model is reduced by 21.19% with the incorporation of signal decomposition. These results underscore the pivotal role of weather parameters and signal decomposition in enhancing the accuracy of electricity load forecasting.

The results highlight the possibility of integrating geographically informed weather selection with frequency domain signal processing to enhance the accuracy and resilience of short-term load forecasting models, which are crucial for load management and operational decision-making in power systems. The suggested method demonstrates

universal applicability. However, regional peculiarities and data availability may influence its effectiveness. Consequently, further research should explore adaptive spatial modelling and real-time data assimilation to enhance prediction dependability across diverse power system scenarios.

CRedit authorship contribution statement

Indra A. Aditya: Writing – original draft, Visualization, Software, Resources, Project administration, Investigation, Formal analysis, Data curation. **Didit Adytia:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors have declared no conflict of interest

Acknowledgements

This research was funded by PT PLN Research Institute, Indonesia with contract number: 0975/SDM.02.07/F30000000/2023.

Data availability

The authors do not have permission to share data.

References

- [1] Wang J, Azam W. Natural resource scarcity, fossil fuel energy consumption, and total greenhouse gas emissions in top emitting countries. *Geosci Front* 2023;101757.
- [2] United Nations Environment Programme, International Resource Panel. Green energy choices: The benefits, risks and trade-offs of low-carbon technologies for electricity production - summary for policy makers. 2015, URL <https://wedocs.unep.org/20.500.11822/7555>.
- [3] Qubeissi MA, El-kharouf A, Soyhan HS. Renewable energy. Rijeka: IntechOpen; 2020, <http://dx.doi.org/10.5772/intechopen.81765>.
- [4] Li L, Ota K, Dong M. When weather matters: IoT-based electrical load forecasting for smart grid. *IEEE Commun Mag* 2017;55(10):46–51.
- [5] Gao S, Liu Y, Wang J, Wang Z, Wenjun X, Yue R, et al. Short-term residential load forecasting via transfer learning and multi-attention fusion for EVs' coordinated charging. *Int J Electr Power Energy Syst* 2025;164:110349.
- [6] Salam A, El Hibaoui A. Energy consumption prediction model with deep inception residual network inspiration and LSTM. *Math Comput Simulation* 2021;190:97–109.
- [7] Haiges R, Wang Y, Ghoshay A, Roskilly A. Forecasting electricity generation capacity in Malaysia: an auto regressive integrated moving average approach. *Energy Procedia* 2017;105:3471–8.
- [8] Dudek G. Pattern-based local linear regression models for short-term load forecasting. *Electr Power Syst Res* 2016;130:139–47.
- [9] Bouktif S, Fiaz A, Ouni A, Serhani MA. Optimal deep learning lstm model for electric load forecasting using feature selection and genetic algorithm: Comparison with machine learning approaches. *Energies* 2018;11(7):1636.

- [10] Hochreiter S. Long short-term memory. *Neural Comput* 1997.
- [11] Yang A, Li W, Yang X. Short-term electricity load forecasting based on feature selection and least squares support vector machines. *Knowl-Based Syst* 2019;163:159–73.
- [12] Li S, Goel L, Wang P. An ensemble approach for short-term load forecasting by extreme learning machine. *Appl Energy* 2016;170:22–9.
- [13] Zhang J, Wei Y-M, Li D, Tan Z, Zhou J. Short term electricity load forecasting using a hybrid model. *Energy* 2018;158:774–81.
- [14] Thissen U, Van Brakel R, De Weijer A, Melssen W, Buydens L. Using support vector machines for time series prediction. *Chemometr Intell Lab Syst* 2003;69(1–2):35–49.
- [15] Box GE, Pierce DA. Distribution of residual autocorrelations in autoregressive-integrated moving average time series models. *J Amer Statist Assoc* 1970;65(332):1509–26.
- [16] Zhang J, Walter G, Miao Y, Lee WNW. Wavelet neural networks for function learning. *IEEE Trans Signal Process* 1995;43(6):1485–97. <http://dx.doi.org/10.1109/78.388860>.
- [17] Ahmad A, Xiao X, Mo H, Dong D. TFTformer: A novel transformer based model for short-term load forecasting. *Int J Electr Power Energy Syst* 2025;166:110549.
- [18] Liu M, Xia C, Xia Y, Deng S, Wang Y. TDCN: A novel temporal depthwise convolutional network for short-term load forecasting. *Int J Electr Power Energy Syst* 2025;165:110512.
- [19] Dong J, Jiang Y, Chen P, Li J, Wang Z, Han S. Short-term power load forecasting using bidirectional gated recurrent units-based adaptive stacked autoencoder. *Int J Electr Power Energy Syst* 2025;165:110459.
- [20] Zhao P, Hu W, Cao D, Zhang Z, Liao W, Chen Z, et al. Enhancing multivariate, multi-step residential load forecasting with spatiotemporal graph attention-enabled transformer. *Int J Electr Power Energy Syst* 2024;160:110074.
- [21] Arvanitidis AI, Bargiotas D, Kontogiannis D, Fevgas A, Alamaniotis M. Optimized data-driven models for short-term electricity price forecasting based on signal decomposition and clustering techniques. *Energies* 2022;15(21):7929.
- [22] Zhou Y, Wang J, Li Z, Lu H. Short-term photovoltaic power forecasting based on signal decomposition and machine learning optimization. *Energy Convers Manage* 2022;267:115944.
- [23] Chen T, Guestrin C. Xgboost: A scalable tree boosting system. In: *Proceedings of the 22nd ACM sigkdd international conference on knowledge discovery and data mining*. 2016, p. 785–94.
- [24] Freund Y, Schapire RE, et al. Experiments with a new boosting algorithm. In: *Icml*, vol. 96, Citeseer; 1996, p. 148–56.
- [25] Taud H, Mas J-F. Multilayer perceptron (MLP). In: *Geomatic approaches for modeling land change scenarios*. Springer; 2018, p. 451–5.
- [26] De Felice M, Alessandri A, Ruti PM. Electricity demand forecasting over Italy: Potential benefits using numerical weather prediction models. *Electr Power Syst Res* 2013;104:71–9.
- [27] Dehalwar V, Kalam A, Kolhe ML, Zayegh A. Electricity load forecasting for urban area using weather forecast information. In: *2016 IEEE international conference on power and renewable energy*. IEEE; 2016, p. 355–9.
- [28] Friedrich L, Afshari A. Short-term forecasting of the Abu Dhabi electricity load using multiple weather variables. *Energy Procedia* 2015;75:3014–26.
- [29] Son H, Kim C. Short-term forecasting of electricity demand for the residential sector using weather and social variables. *Resour Conserv Recycl* 2017;123:200–7.
- [30] Elamin N, Fukushima M. Modeling and forecasting hourly electricity demand by SARIMAX with interactions. *Energy* 2018;165:257–68.
- [31] Wang Z, Tian C, Zhu Q, Huang M. Hourly solar radiation forecasting using a volterra-least squares support vector machine model combined with signal decomposition. *Energies* 2018;11(1):68.
- [32] Feng Z-k, Niu W-j, Wan X-y, Xu B, Zhu F-l, Chen J. Hydrological time series forecasting via signal decomposition and twin support vector machine using cooperation search algorithm for parameter identification. *J Hydrol* 2022;612:128213.
- [33] An N, Zhao W, Wang J, Shang D, Zhao E. Using multi-output feedforward neural network with empirical mode decomposition based signal filtering for electricity demand forecasting. *Energy* 2013;49:279–88.
- [34] Jaseena KU, Koor BC. Decomposition-based hybrid wind speed forecasting model using deep bidirectional LSTM networks. *Energy Convers Manage* 2021;234:113944.
- [35] da Silva RG, Ribeiro MHD, Moreno SR, Mariani VC, dos Santos Coelho L. A novel decomposition-ensemble learning framework for multi-step ahead wind energy forecasting. *Energy* 2021;216:119174.
- [36] Zhang Z, Wang Y, Wang K. Fault diagnosis and prognosis using wavelet packet decomposition, Fourier transform and artificial neural network. *J Intell Manuf* 2013;24:1213–27.
- [37] Duhamel P, Vetterli M. Fast Fourier transforms: a tutorial review and a state of the art. *Signal Process* 1990;19(4):259–99.
- [38] González S, García S, Del Ser J, Rokach L, Herrera F. A practical tutorial on bagging and boosting based ensembles for machine learning: Algorithms, software tools, performance study, practical perspectives and opportunities. *Inf Fusion* 2020;64:205–37.
- [39] Zhou Z-H. *Machine learning*. Springer Nature; 2021.
- [40] Natekin A, Knoll A. Gradient boosting machines, a tutorial. *Front Neuroinformatics* 2013;7:21.
- [41] Bentéjac C, Csörgő A, Martínez-Muñoz G. A comparative analysis of gradient boosting algorithms. *Artif Intell Rev* 2021;54:1937–67.
- [42] Zhang D, Gong Y. The comparison of LightGBM and XGBoost coupling factor analysis and prediagnosis of acute liver failure. *IEEE Access* 2020;8:220990–1003.
- [43] Chen M, Liu Q, Chen S, Liu Y, Zhang C-H, Liu R. XGBoost-based algorithm interpretation and application on post-fault transient stability status prediction of power system. *IEEE Access* 2019;7:13149–58.
- [44] Chen T, He T, Benesty M, Khotilovich V, Tang Y, Cho H, et al. Xgboost: extreme gradient boosting. 2015, p. 1–4, R package version 1102 0.4-2 1 (4).
- [45] Tarwidi D, Pudjaprasetya SR, Adytia D, Apri M. An optimized XGBoost-based machine learning method for predicting wave run-up on a sloping beach. *MethodsX* 2023;10:102119.
- [46] Asuero AG, Sayago A, González A. The correlation coefficient: An overview. *Crit Rev Anal Chem* 2006;36(1):41–59.
- [47] Hodson TO. Root mean square error (RMSE) or mean absolute error (MAE): When to use them or not. *Geosci Model Dev Discuss* 2022;2022:1–10.
- [48] De Myttenaere A, Golden B, Le Grand B, Rossi F. Mean absolute percentage error for regression models. *Neurocomputing* 2016;192:38–48.
- [49] Hersbach H, Bell B, Berrisford P, Hirahara S, Horányi A, Muñoz-Sabater J, et al. The ERA5 global reanalysis. *Q J R Meteorol Soc* 2020;146(730):1999–2049.